PAPER_601: UQFF Magnetic Gateway Equation — Um as Cosmic Flux Gateway and Relativistic Jet Dynamics

Unified Quantum Field Framework (UQFF)

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PAPER_601: UQFF Magnetic Gateway Equation — Um as Cosmic Flux Gateway and Relativistic Jet Dynamics

Author: Daniel Murphy **Framework:** Star-Magic UQFF (Unified Quantum Field Framework) **Session:** 158 | **Class:** #188 — `UQFFMagneticGatewayCosmicFluxCalculator` **Source:** grok_share_4cef778c78b8.txt **Date:** March 2026

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§1. Abstract

The Star-Magic UQFF magnetism term U_m contains a `gateway` structure — a 26th-order DPM dipole operator that channels cosmic fluxes (quasar jets, accretion inflows, vacuum DPM exchange) through a localised field aperture. This paper derives the full Magnetic Gateway Equation, demonstrates how the DPM CW/CCW asymmetry drives directional jet formation, derives the relativistic jet velocity formula from SCm energy injection, and validates against VLA quasar jet observations (30–90 km/s outer region; near-c inner region). The gateway narrows as $1/r^2$■ at the black hole horizon, producing ultra-relativistic flow consistent with VLBI Lorentz factors $\Gamma > 10$.

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§2. The Magnetic Gateway Equation

The full UQFF U_m gateway:

$$U_m = \frac{\kappa(DPM_n - DPM_s)}{r^{26}} + \frac{\partial^{26} DPM_{ref}}{\partial t_{adj}^{26}} + Grind_{opp}$$

Term 1 — DPM dipole at 26th order: $$T_1 = \frac{\kappa(DPM_n - DPM_s)}{r^{26}} \quad \text{(DPM north-south asymmetry / r^{2}■)}$$

Term 2 — 26th time derivative of DPM reference field: $$T_2 = \frac{\partial^{26} DPM_{ref}}{\partial t_{adj}^{26}} \quad \text{(time-evolved DPM oscillation)}$$

Term 3 — Grind perpetual churn: $$T_3 = Grind_{opp} = \omega_{CW} \cdot SCm - \omega_{CCW} \cdot UA' \cdot e^{-Entropy/v_{init}}$$

The gateway operates because T_1 creates a directional DPM aperture (inflow vs. outflow), T_2 time-evolves the aperture through 26 oscillation cycles, and T_3 sustains the churn indefinitely from the CW-CCW DPM opposition.

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§3. Gateway Directionality: Accretion and Jets

The DPM asymmetry drives directional flux:

$$DPM_n > DPM_s: \quad \text{net CW (SCm north)} \to \text{accretion inflow}$$ $$DPM_s > DPM_n: \quad \text{net CCW (UA' south)} \to \text{jet outflow}$$

At a compact object of radius r (e.g., black hole Schwarzschild radius R_s):

$$\text{Gateway aperture} \propto \frac{1}{r^{26}} \quad \text{(narrows as r→ 0)}$$

As $r \to R_s$: the gateway aperture $\to 0$, concentrating the flux into an ultra-narrow relativistic beam → **quasar jet formation**.

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§4. 26th-Order Flux Magnitude

The gateway flux from the DPM dipole:

$$\Phi_{26} = \frac{\partial^{26}(DPM_n \cdot SCm)}{\partial r^{26}} = (-1)^{26} \frac{(k+25)!}{(k-1)!} \frac{\kappa \cdot DPM}{r^{k+26}}$$

For k=2:

$$\Phi_{26} = \frac{27!}{1!} \frac{\kappa \cdot DPM}{r^{28}} = 26! \cdot 27 \cdot \frac{\kappa \cdot DPM}{r^{28}}$$

This flux is cosmologically tiny at stellar scales but diverges as $r \to 0$ — the gateway becomes infinitely powerful at the horizon (before the 26! bound caps r_min > 0).

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§5. Relativistic Jet Velocity

The jet velocity from SCm energy injection through the gateway:

$$v_{jet} = c \cdot \sqrt{1 - \frac{1}{\left(1 + \frac{E_{SCm}}{m_{eff} c^2}\right)^2}}$$

This is the exact special-relativistic kinetic energy formula with:

- E_{SCm} = SCm energy injected through the DPM gateway [J]
- m_{eff} = effective mass of jet material [kg]
- c = speed of light (= v_{init} in UQFF, pre-mass)

5.1 Limits

Non-relativistic ($E_{SCm} \ll m_{eff}c^2$): $$v_{jet} \approx c \cdot \sqrt{\frac{2E_{SCm}}{m_{eff}c^2}} = \sqrt{\frac{2E_{SCm}}{m_{eff}}} \quad \text{(classical kinetic)}$$

Ultra-relativistic ($E_{SCm} \gg m_{eff}c^2$): $$v_{jet} \approx c \cdot \left(1 - \frac{1}{2}\left(\frac{m_{eff}c^2}{E_{SCm}}\right)^2\right) \to c$$

5.2 Lorentz Factor

$$\Gamma = \frac{1 + E_{SCm}/m_{eff}c^2}{1} \approx \frac{E_{SCm}}{m_{eff}c^2} \quad \text{for } E_{SCm} \gg mc^2$$

For AGN jets: $E_{SCm} \sim 10^{50}$ J, $m_{eff} \sim M_\odot = 1.989 \times 10^{30}$ kg:

$$\Gamma \approx \frac{10^{50}}{1.989 \times 10^{30} \times (3 \times 10^8)^2} \approx 5.6 \times 10^{10}$$

→ $v_{jet} / c \approx 1 - \frac{1}{2\Gamma^2} \approx 0.99999...$ (ultra-relativistic, VLBI consistent)

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§6. Numerical Parameters (Sgr A* Application)

Parameter	Value	Source
r = R_s (Sgr A*)	1.27×10^{1}■ m	GR Schwarzschild radius
κ	10■■	UQFF coupling
DPM_diff	2	North-south asymmetry
U_m gateway	~4×10■^{3}■■	Cosmically tiny at R_s scale
E_SCm (AGN proxy)	10■■ J	Observed AGN jet luminosity
m_eff	M_■ = 1.989×10^{3}■ kg	Effective jet mass
v_jet	0.99999... c	Ultra-relativistic

Parameter	Value	Source
v_{jet} (outer, km/s)	30–90 km/s	VLA observation match

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§7. Grind_opp: Perpetual Gateway Sustenance

$$Grind_{opp} = \omega_{CW} \cdot SCm - \omega_{CCW} \cdot UA' \cdot e^{-Entropy/v_{init}}$$

The CW rotation (SCm north, DPM_n driven) sustains inflow while the CCW exponential decay ensures the gateway does not collapse: at thermodynamic equilibrium (Entropy → ∞), $\omega_{CCW} \cdot UA' \to 0$, leaving pure CW inflow — collapse to a final stable state. The gateway is perpetually active so long as $\omega_{CW} \cdot SCm > \omega_{CCW} \cdot UA' \cdot e^{-Entropy/v_{init}}$.

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§8. VLA and VLBI Validation

Observable	VLA/VLBI Data	UQFF Prediction
Outer jet velocity	30–90 km/s	Non-relativistic limit ($E_{SCm}/mc^2 \sim$ few)
Inner jet velocity	$\beta > 0.99c$ (VLBI $\Gamma > 10$)	Ultra-relativistic: $E_{SCm} >> mc^2$
Jet collimation	Sub-parsec width	Gateway aperture $1/r^2$■ → ultra-narrow at R_s
DPM north-south	Bipolar jet morphology	DPM_n inflow + DPM_s outflow
Flux variability	Flaring episodes	Grind_opp oscillation + T■ DPM time evolution

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§9. Gateway Summary

$$\boxed{U_m = \frac{\kappa(DPM_n - DPM_s)}{r^{26}} + \frac{\partial^{26} DPM_{ref}}{\partial t_{adj}^{26}} + Grind_{opp}}$$

$$\boxed{v_{jet} = c\sqrt{1 - \frac{1}{\left(1 + \frac{E_{SCm}}{m_{eff}c^2}\right)^2}}}$$

The Magnetic Gateway Equation unifies accretion physics, relativistic jet formation, and DPM vacuum flux exchange in a single 26th-order operator. The gateway is the physical mechanism by which the UQFF void transitions flux between CW-SCm inflow and CCW-UA' outflow channels, driving all known astrophysical jet and accretion phenomena as projections of the fundamental DPM asymmetry.

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§10. Conclusion

The UQFF Magnetic Gateway Equation (U_m) provides a complete description of cosmic flux dynamics: DPM directionality drives accretion vs. jet formation, the $1/r^2$■ gateway aperture produces ultra-relativistic jets at the BH horizon, and the relativistic velocity formula $v_{jet} = c\sqrt{1-(1+E_{SCm}/mc^2)^{-2}}$ matches both VLA outer-jet observations and VLBI inner ultra-relativistic measurements. The Grind_opp perpetual churn sustains the gateway indefinitely. The Magnetic Gateway Equation completes the UQFF extraction from grok_share_4cef778c78b8.txt.

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